

Date _____

Day _____

Introduction To Data Engineering

→ Airbnb : Price & Availability ^{each}

↳ they stores price & avail of a property for 365 days using ML algorithms

	Property_id	date	price	is_available	...
365 days	7	01-01-2026	10,000	True	..
	7	:	:	}	..
	7	31-12-2026	20,000	False	..
365 days	12	01-01-2026	50,000	True	..
	:	:	:	:	..

Total no. of properties = 6M

Days to store = 365 days

Total Rows = 6M × 365
= 2.2 billion

if no. of properties ↑

↳ 60M → 22 billion

↳ 600M → 220 billion

↳ Too much High volume data

Page No. _____

Date _____

Day _____

! Therefore the processing & storage becomes a problem

↳ Slow Queries

↳ If there are JOINS → more painful

→ Velocity ⇒ speed & freq. at which data is generated, captured, or processed

↳ Batch Processing (Low Velocity) → data is collected over time and processed in chunks (weekly, nightly)

Apache Spark ↳ Micro Batching (Med Velocity) → Data is processed in tiny increments every few minutes or seconds

↳ Streaming / Real Time (High) → Data is processed instantly. Ex: fraud detection, live IoT sensor's feed, stock market

↳ Kafka, ~~A~~

→ Variety → Different types of data

↳ Text, Images, sensor-data, JSON logs, XML Binary

↳ 3rd party integrations have their own formats

Page No. _____

Date _____

Day _____

→ To cater these problems of storage, velocity & variety of data, new technologies were introduced

- Dataware houses

- ↳ Central repository for structured data collected from various operational sys (CRMs, ERP, etc)

- ↳ Uses concept "Schema On Write"

- ↳ Before data enters, it must be cleaned, transformed, & fitted into a rigid structure of tables

- ↳ Optimized for analytics

- ↳ Stores data Column-by-Column

- ↳ Snowflake, Google BigQuery, Amazon Redshift

- Data Lakes

- ↳ A vast pool that stores data in its natural, raw state.

- ↳ Can hold structured, semi-structured, or unstructured data

- ↳ Used concept "Schema On Read"

- ↳ Dump data immediately without processing it

- ↳ Amazon S3, Azure Blob Storage, Google Cloud Storage

Page No. _____

Date _____

Day _____

→ Veracity ⇒ refers to the accuracy, trustworthiness and reliability of data

↳ Focus is on quality of data

↳ Low veracity data

↳ Typos, missing fields, user entering fake info

↳ duplicate records

→ 4Vs Of Data Engineering

• Velocity • Volume • Variety • Veracity

→ OLTP Vs OLAP

↳ These are the two core database workload categories

OLTP

1. Online Transaction Processing
2. Primary purpose is day-to-day operations (eg: recording a sale or transacting funds)

OLAP

1. Online Analytical Processing
2. Generating reports, identifying trends and decision-making

Page No. _____

→ We always anonymize the PII (Personal Identity Info)
in OLAP. R

Date _____

Day _____

3. Real-time, current operational data	3. Historical data consolidate from multiple sources
4. Row-oriented	4. Column-oriented
5. Low Latency, Low Volume	5. High Latency, High Volume
6. JOINS are OKAY ↳ Normalization ↳ No duplication	6. JOINS are NOT GOOD ↳ denormalization
7. Resp time → ms	7. Resp time → secs to min
8. MySQL, PostgreSQL, ORACLE	8. Snowflake, Google Big Query

↳ Columnar Database:

↳ A data management system that stores data by column instead of rows

↳ Compression Of Data (Parquet)

compressed [200, 200, 200, 192, 192, 302, 302]

↳ [200, NULL, NULL, 192, NULL, 302, NULL]

↳ repetitive data par NULL rakhdia.

↳ Even 20% compression of Peta Bytes data saves millions of dollars

Date _____

Day _____

→ For denormalization, we can make a log table of Events and then transfer the data of the normalized tables into that event table.

→ But, denormalization alone doesn't solve problems if we have 3 billion DAU,

↳ 10 events/day/user → 30 billion log records

↳ Per year → 30 Billion × 365

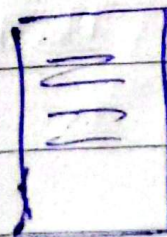
→ 10.950×10^9

→ 10 TB =

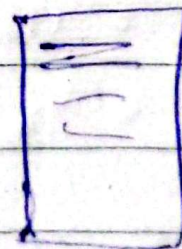
↳ query on 10TB record is also painful

↳ Now to solve this we partition the data into days & every day we process for that partition & stores the result in a.

Fact Table



10/02/26



9/02/26

Date _____

Day _____

→ So now at the end of year we would have only 365 records in FACT Table.

→ Airflow

↳ Used to programmatically write, schedule and monitor complex workflows.

↳ DAGs: In Airflow, a workflow is defined as a Directed Acyclic Graph.

↳ A collection of task you want to run.

↳ Tasks: Individual units of work in the pipeline
eg: Pulling data from an API,
Running a SQL Query,
Training a ML model

→ AWS Step Function can also be used for this purpose

Date _____

Day _____

• Usecase : IoT System Design

- 1 Million homes
 - On avg : 50 IoT nodes / home
 - Freq of data to cloud : 5 secs
- ↳ node mean any electronic appliance (AC, Fridge, TV)

Find the time to process

↳ day by day data ↳ Month by Month data ↳ Year by Year data

↳ Table → home-id - device-id - energy-consumption

→ Total Nodes = $1M \times 50 \text{ devices} = 50 M$

→ Data transferred to cloud in 5 sec → 50 M
1 sec → 10 M

QPS = $10M \text{ msgs/sec}$
↳ Traffic estimation

Storage Estimation

↳ per day = $10M \times (60 \text{ sec} \times 60 \text{ min} \times 24 \text{ hrs})$
= $10M \times 86400$
= 864×10^9
= 864 billion msgs/day

Page No. _____

or rows/day

Date _____

Day _____

Storage (120 B) $\rightarrow$ one row

$$\begin{aligned}\hookrightarrow \text{per sec} &= 10M \times 120B \\ &= 1200 MB/s \\ &= 1.2 GB/s\end{aligned}$$

$$\begin{aligned}\hookrightarrow \text{Per day} &= 1.2 GB/s \times 86400 \\ &= \boxed{100 TB / \text{day}}\end{aligned}$$

$$\begin{aligned}\hookrightarrow \text{Per month} &= 100 TB / \text{day} \times 30 \\ &= \boxed{3000 TB / \text{month}}\end{aligned}$$

$$\begin{aligned}\hookrightarrow \text{Per Year} &= 300 TB / \text{month} \times 12 \\ &= \boxed{3600 TB / \text{year}}\end{aligned}$$

$\rightarrow$ On Avg. a production grade PostgreSQL can process 3M rows/sec

$$\begin{aligned}\bullet \text{ day by day} &= \frac{864 \times 10^9}{3 \times 10^6} = 288,000 \text{ sec} \\ &\hookrightarrow \boxed{80 \text{ hrs}}!\end{aligned}$$

$$\bullet \text{ Month by Month} = 80 \times 30 = 2400 \text{ hrs}$$

$$\bullet \text{ Yr by Yr} = 2400 \times 12 = 24,000 \text{ hrs}$$

OLTP Will Fail!

Page No. _____

Date _____

Day _____

• Solution Q1

↳ Convert Raw data $\xrightarrow{80 \text{ hrs}}$ Day by day fact table
↳ Partitioned into day by day

day by day fact of each device

we get a SOM rows table

SOM = 16 sec

Now we have

Month by month fact table

SM

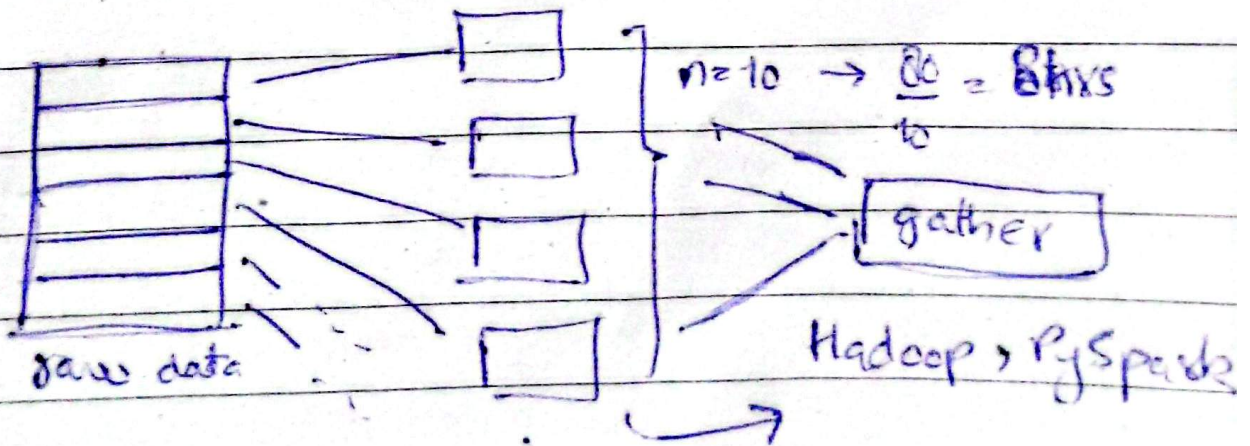
to only process

SOM rows for month by month

↳ has dim month by month k like sun k range or iteratively aggregation kote change. Rozana 16 sec ki query chalogi

Year by Year fact table

→ To Reduce the 80 hrs for day by day processing we can do parallel processing.



MQTT: Message Queuing Telemetry Transport. It is a lightweight open-standard messaging protocol designed specifically for machine to machine (M2M) communication and IoT

Date _____

Day _____

